Review of "Natural Capital and the Measurement of Productivity"

This paper examines the possibility that the exclusion of natural capital services from standard growth accouting biases estimates of total factor productivity growth. It does this by looking at how correlated stocks of capital (which are presumed to be proportional to flows) are to existing measures of TFP growth, and finds meaningful correlations, which are suggestive of bias.

A broad question

One of the things I'm worried about is the presumption that many of the unmeasured natural services mentioned (e.g. nutrient cycling in fisheries) are incorporated somehow in GDP growth, and hence in the "true" measure of TFP, but are unmeasured as inputs, inflating/deflating the Solow residual. It is not obvious that these services end up influencing GDP in the first place. To use the nutrient cycling example, for that service to be incorporated into GDP it must be that the value of it shows up in the price of the output - fish - that are caught and sold. I might believe that the fishing industry internalizes the effect of their catch on future stocks of fish - we have licenses and catch limits for a reason - but it seems hard to believe that they internalize the impact of their catch on the nutrient cycling that occurs because of fisheries and which then influences the health of other fish, ocean plants, and terrestial plants indirectly. Because they don't internalize this, the effect of nutrient cycling is not included in the price of fish. Because the price of fish doesn't reflect the value of nutrient cycling, GDP doesn't reflect the value of nutrient cycling. And because GDP doesn't reflect the value of nutrient cycling, but that nutrient cycling does have a physical impact on production in some way, that has to show up in TFP (where else would it show up?).

I think what I'm trying to say is that your identification assumption that "true" TFP growth is uncorrelated to changes in the stock of environmental capital (and hence with changes in the flow of their services) has to be wrong? We have to be careful here, because there is no such thing as a pristine actual true handed-down-by-God growth rate to TFP. TFP growth is always and forever only a residual, and as such it depends on the measurement of both inputs *and* the measurement of GDP. So what you are calling "true TFP" growth is still in fact picking up environmental service flows because GDP is *not*. And hence it must be correlated with your natural capital stocks?

In some sense this is great for you, because it means that you don't care why natural capital stocks are correlated with the Solow residual. Either it is because $\gamma_X > 0$ or because TFP is picking up unpriced natural service flows. It makes your regression results more plausible (I'm less worried they are spurious) but at the same time less informative (natural capital matters, we just don't know how).

A procedural question

The biggest concern I have is the assumption that the elasticities, γ_x, are invariant over time and across countries. There is no question this is wrong. Now, perhaps one can argue that it isn't wrong in a way that really matters (the variation is small on both dimensions). But if we're arguing that small differences in γ_X don't matter, and you end up assuming that $\gamma_X = 0.05$ (as in section 4), I have to ask why I care whether we included or exclude natural capital when calculating TFP?

Your initial setup of regressing measured Solow residuals on natural capital stocks makes sense (even

taking into account my first point). That approach tells us something about whether natural capital belongs in the accounting or not. Full stop.

But to know how *much* it matters to the accounting, you have to take a stand on the size of γ_x. The regressions cannot give us this, as you demonstrate. So immediately we are back to the doing something which you were (rightly) trying very hard to avoid in the main paper, which is "regress output on natural capital stocks and see what you get". Equation (17) just does the thing we always say not to do, which is to try and estimate a production function by regressing output on inputs. And sure, you have some controls, but we can't take the results too seriously because not matter what, we can't be sure that unobserved shocks to productivity (epsilon) aren't correlated with the growth rates of renewable and non-renewable resources.

That gives us Table 14, which is then used to justify the "Well, let's say it's about 0.05" assumption for γ_X. If we could get the values of γ_X from the regressions in Table 14, then *why* did you write the paper? If we need the very careful structure and approach of the paper to even hope to understand *if* γ_X *might* be non-zero, then how can we trust Table 14? It's also the case that you can't just add up the estimates in Table 14 to get the uber-elasticity for all resources, because you can't just add up those stocks of resources.

You'll see below that I don't think the use of Table 14 is even necessary, as the whole bias/RMSE thing is overcomplicating things. My suggestion is that you stick with the carefully made point in the early part of the paper, which is that we can get some information about whether natural capital belongs in the accounting by looking at whether natural capital is correlated with existing measures of the Solow residual.

A question on why section 4 exists

I fail to see why I care about the RMSE of Models 1 and 2? I certainly may care about how biased they are, but whether they happen to be noisy or not is just a feature of the data. These are accounting statements, not empirical models, in the end, and I just want to get them right, hence I can want the bias to be low. But we aren't trying to fit an explanatory model for TFP here, we're trying to do accounting, so whether the "fit" is good or not is irrelevant. If once we account for natural capital, the "true" TFP growth rate is noisier than we had assumed, okay. It's noisier than we assumed. That's informative about the true nature of TFP, not a problem.

This might have been a side point, but the paper states "Proposition 2 presents the non-monotonicity result that motivates this paper." Really? In what way? In reading the introduction and earlier parts of the paper I never once internalized the idea that we're trying to reduce the RMSE of the series of TFP growth rates. I did internalize that Solow residuals could be biased because we are failing to measure inputs that should be included. And you're working on that, making some progress. What does Proposition 2 have to do with it?

Which means I don't see the point of section 4.2 at all. The RMSE is *not* a measure of how accurate a series of TFP estimates are, and showing that it goes down doesn't tell me anything. Which you already acknowledged in Proposition 2! Note that in your comparison to existing results on page 22, all those authors are doing is reporting how the estimated growth rate of TFP changed once they accounted for natural capital/resources. Whether that reduced RMSE is beside the point entirely.

Smaller points

1. Footnote 3 worries me. γ_x are elasticities, period. Income shares of factors of production are a different thing, which under very specific - and almost certainly unrealistic - conditions will equal those elasticities. I think the point is just that Section 1 of the Appendix is not just unnecessary, but likely to be wrong. You're already doing the right thing in the main paper.

2. On page 4 in the discussion of how g_A is constructed, I think it is worth noting that the reasons statistical agencies use $I \neq X$ isn't just because they can't observe all the stocks of inputs (natural or otherwise) in X, but because the presumption or measurement of $\gamma_x \approx 0$ for some of them. This is kind of a quibble, but the choice depends on both. It's not worth measuring elements of X if their effect on production is tiny. Now, we may be wrong about that (which is what this paper is about) but the problem isn't simply a lack of ability to measure. It's often a deliberate choice because these inputs are not seen as important. Which gets to the next point.